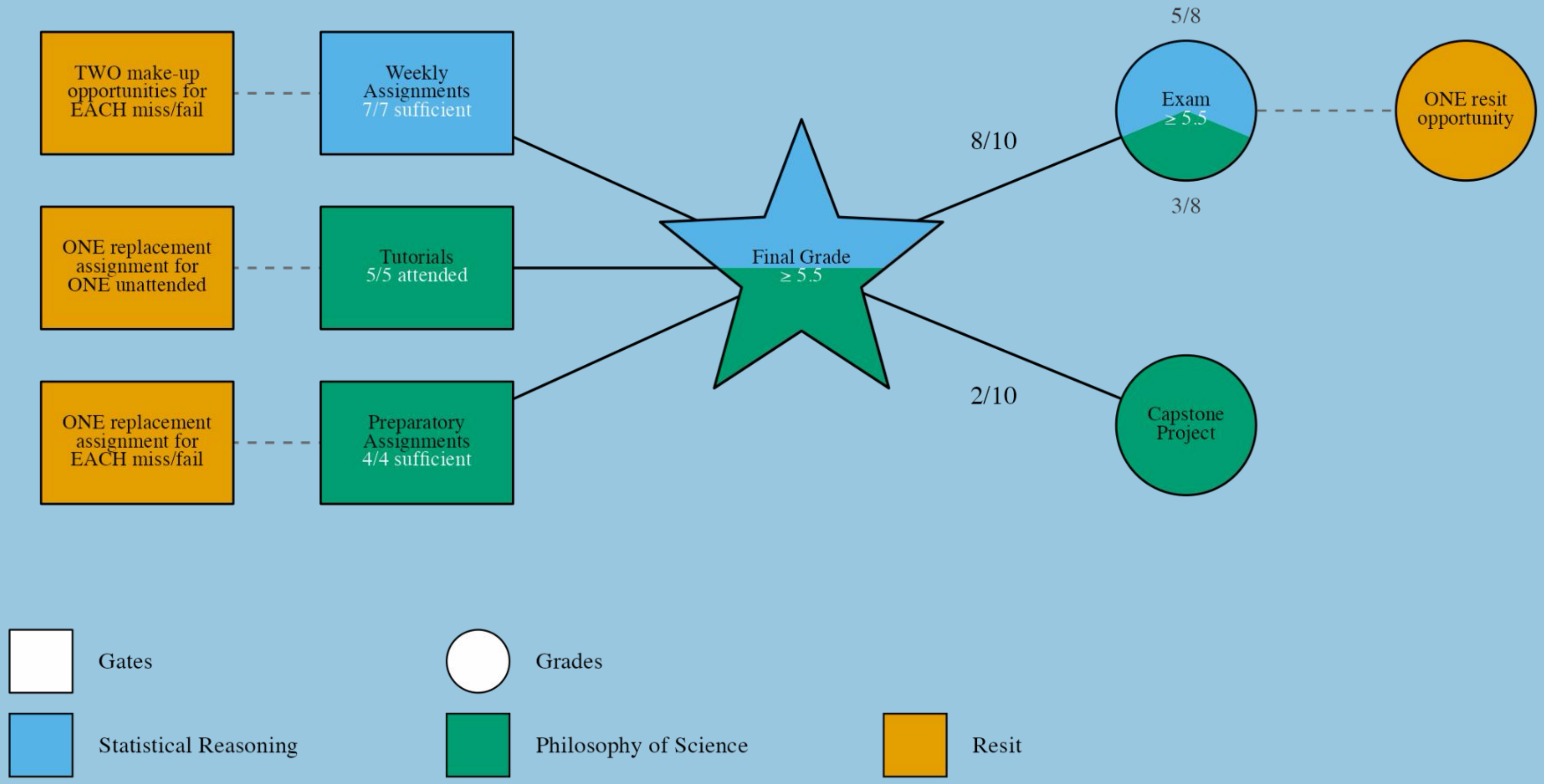
Wetenschapsfilosofie & Statistisch Redeneren

Psychobiologie, Universiteit van Amsterdam

🪢 People

		WSR	WF	SR	💡	🙋
Lecturers	Alexander Savi				Everything cross-course & substantive SR; Jonas first line of contact	Lecture/Canvas
	Jolien Francken				Substantive WF; tutorial teachers first line of contact	Lecture
Teachers	Jonas van Nijnatten				General WF; general SR; weekly assignments; assignment Q&A	Tutorial/Q&A
	Tess den Uyl				General WF	Tutorial
	Madison Carr				“	Tutorial
	Claire Paulissen				“	Tutorial
	Suzanne van Leerdam				“	Tutorial
	Jolande Disser				“	Tutorial
	Eline Hasenoot				“	Tutorial
	Lola van Linge				“	Tutorial
TAs	Maryam Shafik					-
	Nathan Pieterse					-
Education	*Anne/Marleen/etc.*					*Email*

Grading



TWO make-up opportunities for EACH miss/fail
Weekly Assignments 7/7 sufficient
ONE replacement assignment for ONE unattended
Tutorials 5/5 attended
ONE replacement assignment for EACH miss/fail
Preparatory Assignments 4/4 sufficient
Final Grade ≥ 5.5
8/10
2/10
5/8
Exam ≥ 5.5
3/8
ONE resit opportunity
Capstone Project
Gates
Grades
Statistical Reasoning
Philosophy of Science
Resit

Statistical Reasoning

"The most important questions of life are, for the most part, really only problems of probability."
— Pierre-Simon, Marquis de Laplace

👨‍🏫 Statistical Reasoning Lecture #1
Alexander Savi, 2026
📚 Mehmetoglu & Mittner Ch. (1, 2, 3), 7

Curled by Danielle Navarro (jasmines package, diy) 🎨

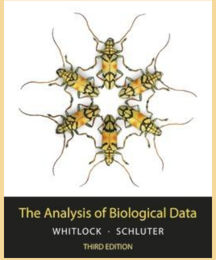
What are you supposed to know already?

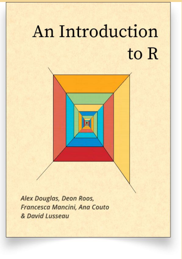
UNIVERSITEIT VAN AMSTERDAM

CANVAS BY INSTRUCTURE

R Studio®

SOWISO



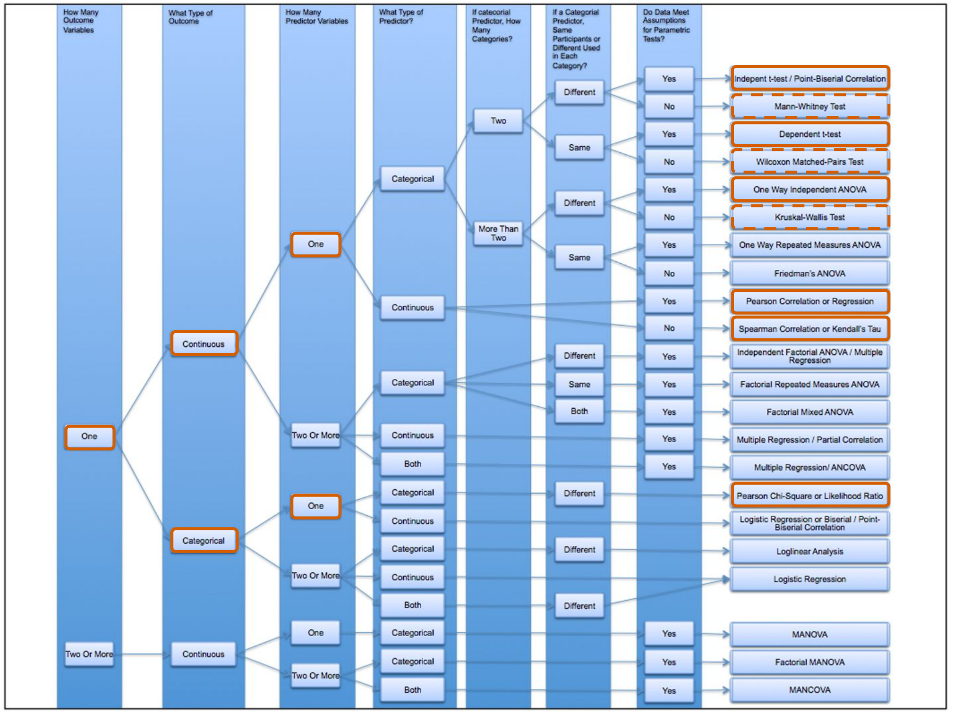


Methoden van Onderzoek en Statistiek

Samples & sampling distributions
NHST
P values
Confidence intervals
Probabilities
Bayes theorem
Test choice & research design
Data visualization

♫ [André van Duin - Statistiek]

What are you supposed to know already?

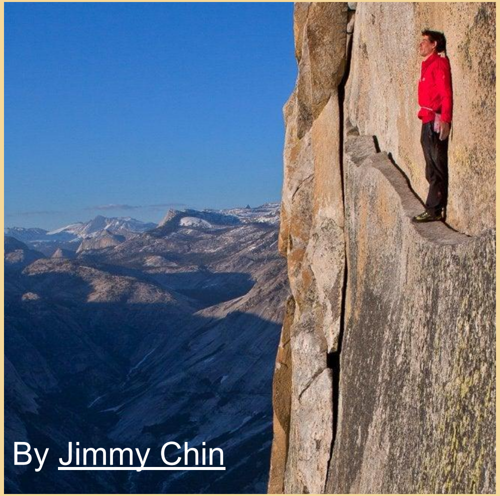


Methoden van Onderzoek en Statistiek

Binomial test
Chi-squared test
Student's *t*-test
One-way ANOVA
Correlation
Simple linear regression
Some nonparametric alternatives

Does the unprescribed use of Ritalin (yes, no) lead to mental illnesses, such as sleeplessness and psychological dependence (yes, no)?

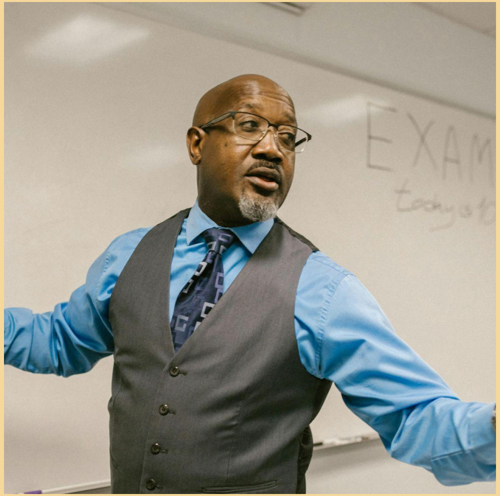
By Bhekisisa.org

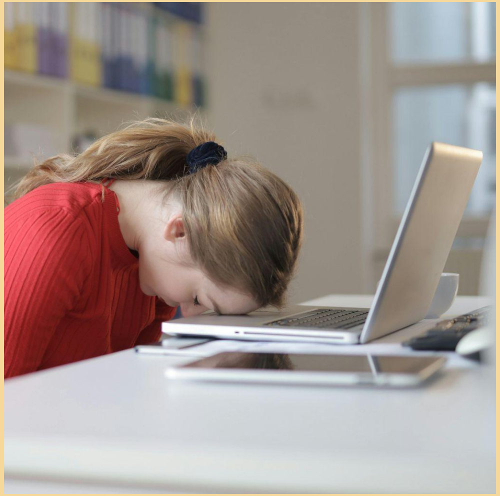


By Jimmy Chin

How does the amount of daily physical activity (measured in hours) predict the likelihood of experiencing anxiety symptoms (yes, no)?

Is there a difference in average test scores among students from different teaching methods (traditional, online, hybrid)?





What is the relationship between hours of sleep per night and cognitive function (measured by memory recall)?

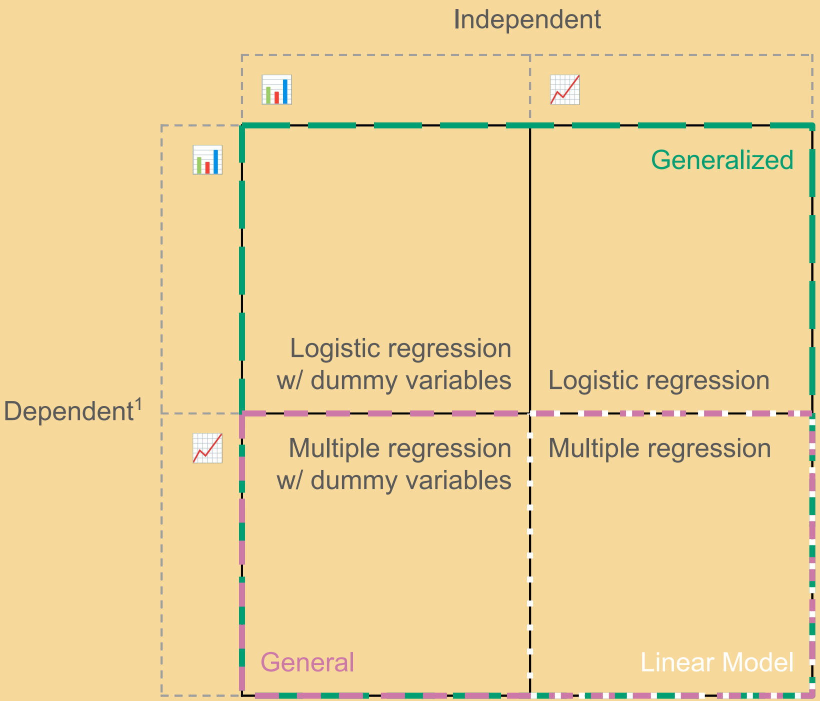
How do we go from here?

Dependent[1] \ Independent	📊	📈
📊	Chi-squared test	
📈	Student's *t* test One-way ANOVA Factorial ANOVA	Simple regression Multiple regression

- ❏ From one predictor to multiple predictors

📊 Categorical: dichotomous[1], polytomous
📈 Continuous: real[1], integer, bounded (e.g., 0 to 1)

How do we go from here?

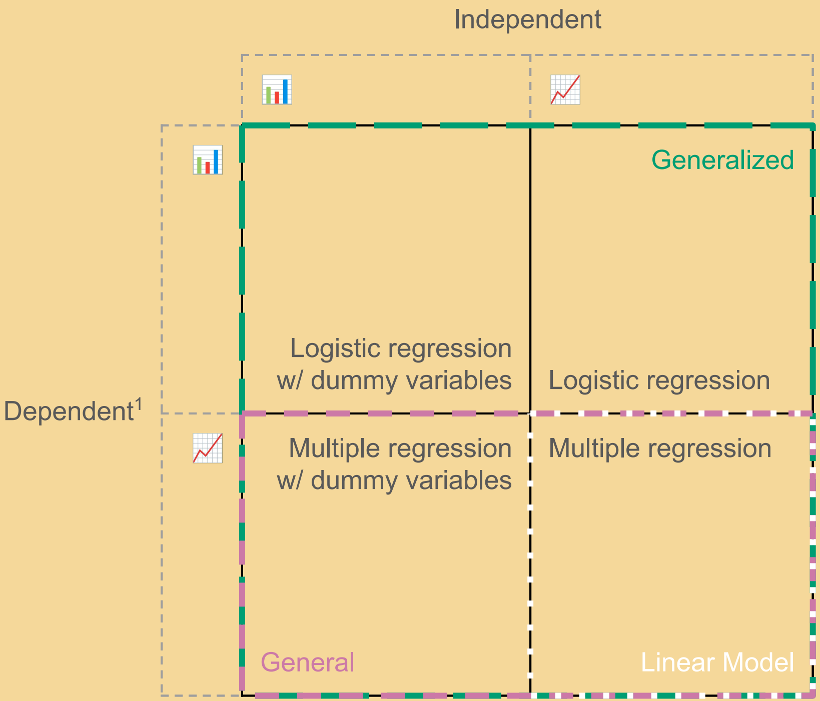


- ❏ From one predictor to multiple predictors
- ❏ From a toolbox to a multitool (“Swiss Army knife”)

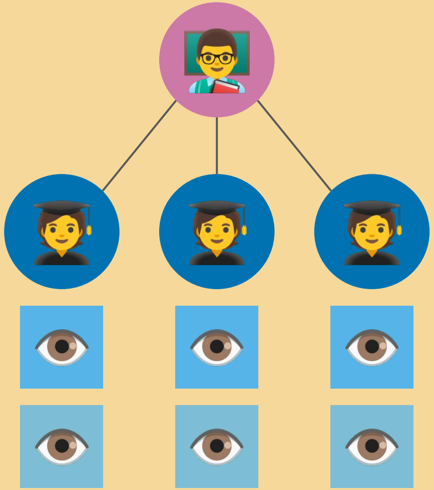
📊 Categorical: dichotomous[1], polytomous
📈 Continuous: real[1], integer, bounded (e.g., 0 to 1)

📷 [Francis Flinch]

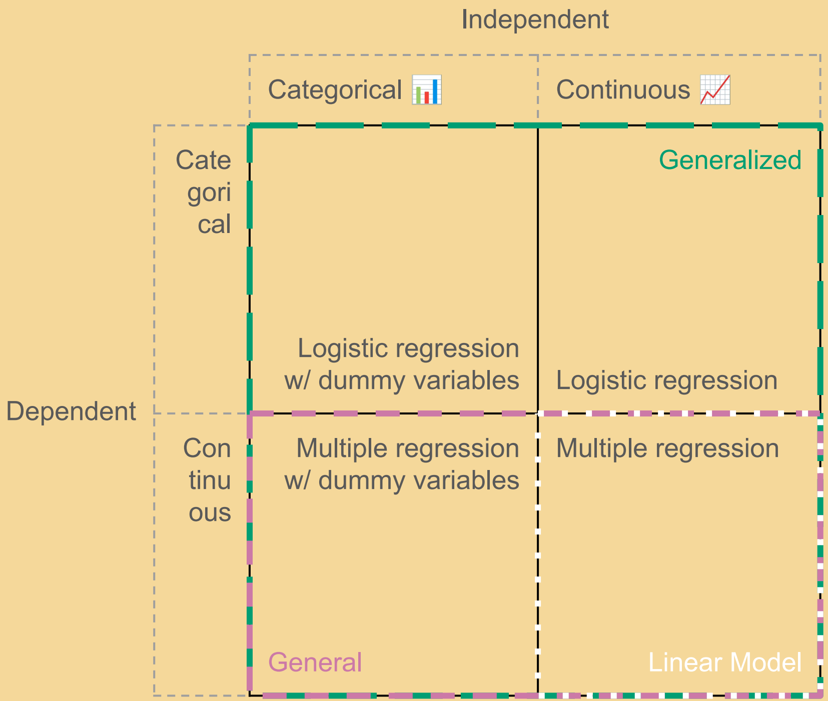
How do we go from here?



- ❏ From one predictor to multiple predictors
- ❏ From a toolbox to a multitool
- ❏ Hierarchical data analysis



How do we go from here?



- ❏ From one predictor to multiple predictors
- ❏ From a toolbox to a multitool
- ❏ Hierarchical data analysis
- 🥷 Bayesian inference (vs. frequentist inference)

Course overview

Curled by Danielle Navarro (jasmines package, diy) 🎨

👨‍🏫 Lectures

🙋 Questions during & after lecture

[Alexander Savi] (he/him)

- ❏ Honours BSc Psychology
- ❏ Research MSc Psychology
- ❏ PhD Psychological Methods
- ❏ Postdoc Learning Analytics
- ❏ Assistant Professor of Psychological Methods

Topics

1. Statistical reasoning with GLM
2. Multiple linear regression (GLM)
3. Dummy-variable regression (GLM)
4. Intermezzo: Bayesian statistics
5. Logistic regression (GLM)
6. Multilevel and longitudinal analysis (GLM)
7. Statistics superpowers

Why attend

- ❏ Prioritized concepts from the book
- ❏ Fresh examples
- ❏ Bare necessary R code
- ❏ Book links for quick reference
- ❏ Topics not covered by the book
- ❏ Some extras for those who want more
- ❏ Ask questions

Assistants

🦸 Miró the Superhero
🦹 Willem the Supervillain

⚠️ Web lectures can fail, use for re-viewing

🧑‍💻 Weekly assignments*

🙋 Online Q&A session (drop-in):
(almost) every Friday 10:00–10:45

Jonas van Nijnatten (he/him) + TAs

- ❏ Four sections
  - ❏ R & Tidyverse
  - ❏ Begrip & Inzicht
  - ❏ Analyse & Interpretatie
  - ❏ Casus
- ❏ Formative
- ❏ Q&A session
  - ❏ prepare questions or use it to work on it
- ❏ Use decimal points (.)
- ❏ Great exam preparation
- ❏ Help improve assignments

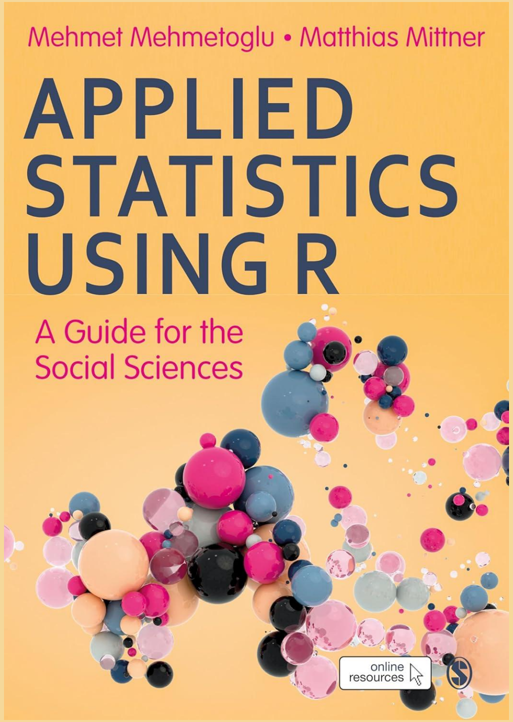
Grading

- ❏ Every Saturday 09:00 to Friday 23:55
- ❏ 80% correct is sufficient, 70% correct depends on shown effort
- ❏ 7/7 sufficient
- ❏ Two attempts per assignment
- ❏ Two catch-up weeks
  - ❏ If sufficient after first: correction to grade
  - ❏ If sufficient after second: correction to retake
- ❏ No feedback; correct answers will be available for a week

* Course manual is leading

⚠️ Recidivists: new Canvas assignments

📚 Book



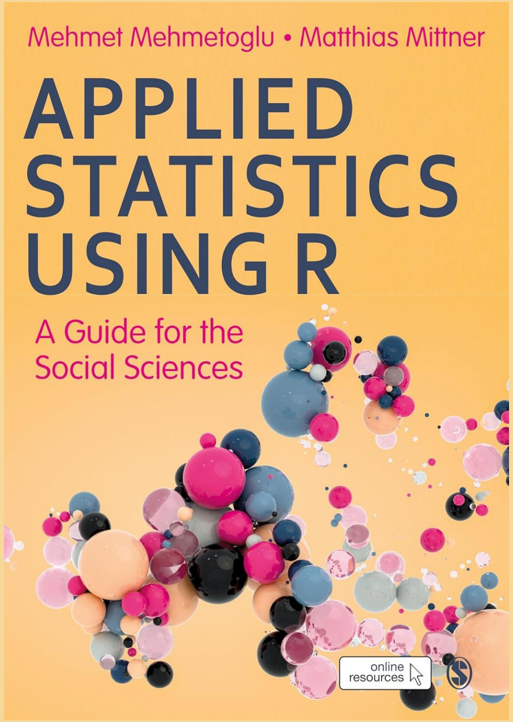
Pros

- ❏ Builds on Chapter 18 of Whitlock & Schluter (where you left of last year)
- ❏ Not only for social sciences
- ❏ Modern approach (GLM and Tidyverse)
- ❏ Frequentist and Bayesian inference, with examples and code
- ❏ Good reference during the remainder of your study
- ❏ It's a book!

Cons

- ❏ It's a book!
- ❏ Some students find it not easy

📚 Book



Active reading

- ❏ Use skimming for some chapters, so you can scan them when there's something you really need to know.
- ❏ Use intensive reading (or 'study') for others (check online resources for executing R code).

Prepare the literature before the start of the week.

💯 Exam*

- ❏ ⅝ SR
- ❏ Digital open book (⚠️)
- ❏ Practice makes perfect
  - ❏ weekly assignments *Begrip & Inzicht* and *Analyse & Interpretatie* sections
  - ❏ exam(ple) questions from previous years
- ❏ Learning goals (studiewijzer)
- ❏ Always do the exam
- ❏ Exam review: Friday, November 13, 9–12

* Course manual is leading / ⚠️ SR only; lecture slides & book chapters as pdf; no weekly assignments; no personal notes

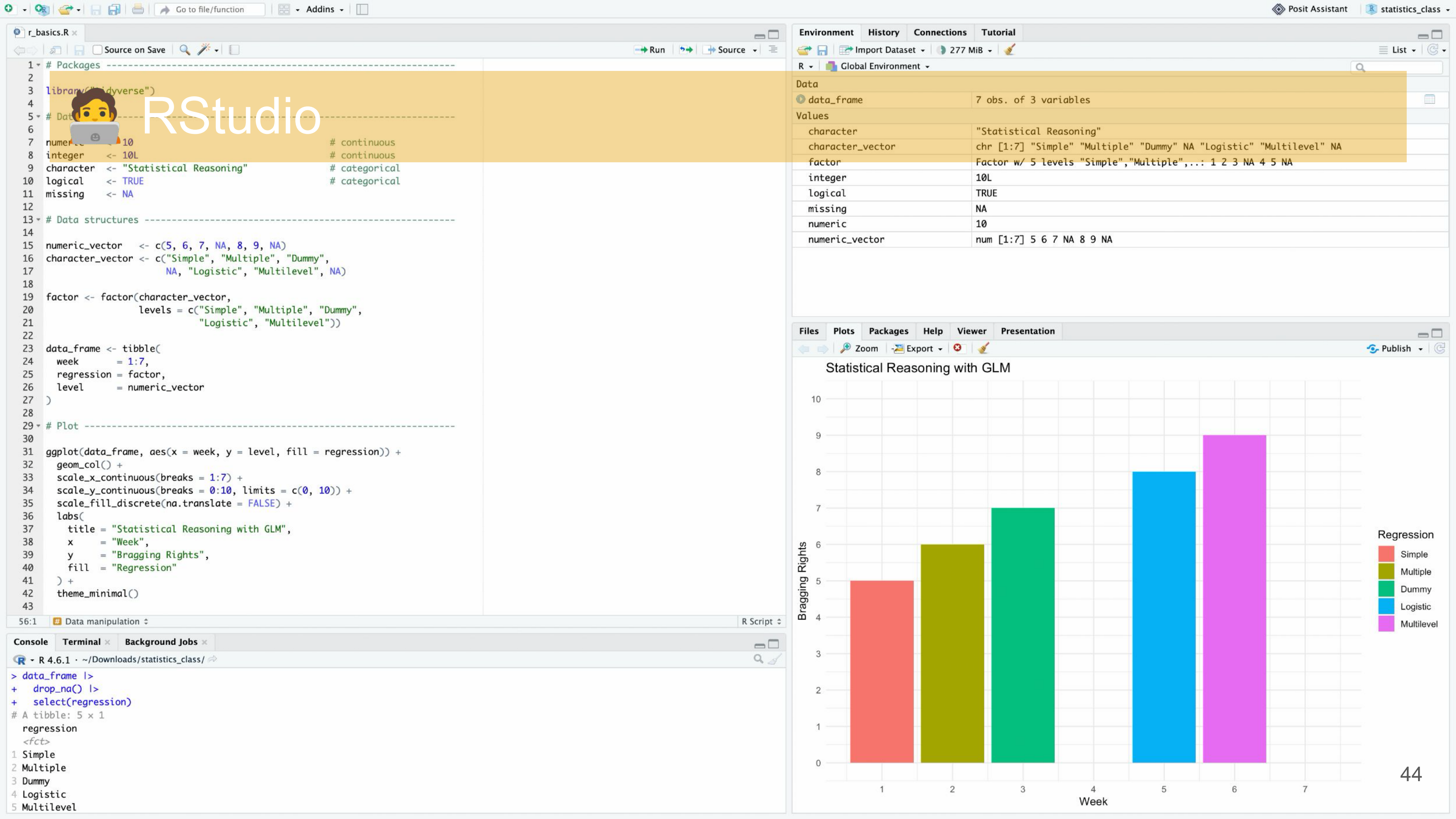
🧑‍💻 R

Statistics for all: open-source pioneers win million-dollar prize

The free, open-source software R has fundamentally transformed research in statistics and data science — and continues to shape the field to this day. Now the pioneers who have driven its development for some 30 years are receiving the Rousseeuw Prize for Statistics,

— [ETH](ETH) (Jun. 17, 2026)

RStudio



r_basics.R
Source on Save
Run
Source
Packages ---
dyverse")
10 # continuous
integer <- 10L # continuous
character <- "Statistical Reasoning" # categorical
logical <- TRUE # categorical
missing <- NA
# Data structures --
numeric_vector <- c(5, 6, 7, NA, 8, 9, NA)
character_vector <- c("Simple", "Multiple", "Dummy",
NA, "Logistic", "Multilevel", NA)
factor <- factor(character_vector,
levels = c("Simple", "Multiple", "Dummy",
"Logistic", "Multilevel"))
data_frame <- tibble(
week = 1:7,
regression = factor,
level = numeric_vector
)
Plot ---
ggplot(data_frame, aes(x = week, y = level, fill = regression)) +
geom_col() +
scale_x_continuous(breaks = 1:7) +
scale_y_continuous(breaks = 0:10, limits = c(0, 10)) +
scale_fill_discrete(na.translate = FALSE) +
labs(
title = "Statistical Reasoning with GLM",
x = "Week",
y = "Bragging Rights",
fill = "Regression"
) +
theme_minimal()
56:1
Data manipulation
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/Downloads/statistics_class/
> data_frame |>
+ drop_na() |>
+ select(regression)
# A tibble: 5 × 1
regression
<fct>
1 Simple
2 Multiple
3 Dummy
4 Logistic
5 Multilevel
Environment
History
Connections
Tutorial
Import Dataset
277 MiB
List
Global Environment
Data
data_frame 7 obs. of 3 variables
Values
character "Statistical Reasoning"
character_vector chr [1:7] "Simple" "Multiple" "Dummy" NA "Logistic" "Multilevel" NA
factor Factor w/ 5 levels "Simple","Multiple",..: 1 2 3 NA 4 5 NA
integer 10L
logical TRUE
missing NA
numeric 10
numeric_vector num [1:7] 5 6 7 NA 8 9 NA
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Statistical Reasoning with GLM
Bragging Rights
Week
Regression
Simple
Multiple
Dummy
Logistic
Multilevel

“Keep the robots out of the gym”

Daniel Miessler; Luke Shepard; Bruce Schneier

UvA AI guide

Logical puzzle | Who owns the zebra?

1. There are five houses.
2. The Englishman lives in the red house.
3. The Spaniard owns the dog.
4. Coffee is drunk in the green house.
5. The Ukrainian drinks tea.
6. The green house is immediately to the right of the ivory house.
7. The Old Gold smoker owns snails.
8. Kools are smoked in the yellow house.
9. Milk is drunk in the middle house.
10. The Norwegian lives in the first house.
11. The man who smokes Chesterfields lives in the house next to the man with the fox.
12. Kools are smoked in the house next to the house where the horse is kept.
13. The Lucky Strike smoker drinks orange juice.
14. The Japanese smokes Parliaments.
15. The Norwegian lives next to the blue house.

In the interest of clarity, it must be added that each of the five houses is painted a different color, and their inhabitants are of different national extractions, own different pets, drink different beverages and smoke different brands of American cigarets [sic]. One other thing: In Statement 6, *right* means *your* right.

—Life International, December 17, 1962

Think

- ❑ Do you **think** you can solve it? (don't **do** it)
- ❑ Why don't you?
- ❑ Why do you?
- ❑ What would be your approach?

Act

- ❑ Solve it (2 minutes, only start)

Reflect

- ❑ Strategies?
- ❑ Collaboration?

"Only 2% of the world population can solve this puzzle."

Who owns the growth mindset?

Similarities

- ❏ Each of you can learn to understand the logic
- ❏ Understanding and confidence builds incrementally
- ❏ Failure can be productive, difficulties can be desirable
- ❏ There are multiple valid strategies
- ❏ Collaboration accelerates progress

Differences

- ❏ The goal is to help you, not puzzle you
- ❏ The skill transfers far beyond this course

Everyone of you can solve it *"by combining deduction, analysis and sheer persistence"*

– Life International (1962)

Single most predictive mindset for succeeding in this course!

📖 Dweck, C. S., & Yeager, D. S. (2019). Mindsets: A view from two eras. *Perspectives on Psychological Science, 14*(3), 481–496.

🔮 # Today

Topics

1 | Statistical reasoning with GLM

  1.1 | 🔁 Simple linear regression

  1.2 | Simulation superpower

2 | Multiple linear regression

3 | Dummy-variable regression

4 | Bayesian statistics (a brief introduction)

5 | Logistic regression

6 | Multilevel and longitudinal analysis

7 | Statistics superpowers

Learning goals

Revisit simple linear regression, which is at the core of this course.

Get acquainted with statistical simulations and see how they can help elucidate statistical concepts.

Revisit some of the concepts of MvO1.

Simple linear regression

Curled by Danielle Navarro (jasmines package, diy) 🎨

IJskoud de Beste

Joop

Can I use the weather forecast to predict the amount of ice cream to make?

What is the relationship between temperature and ice cream sales?

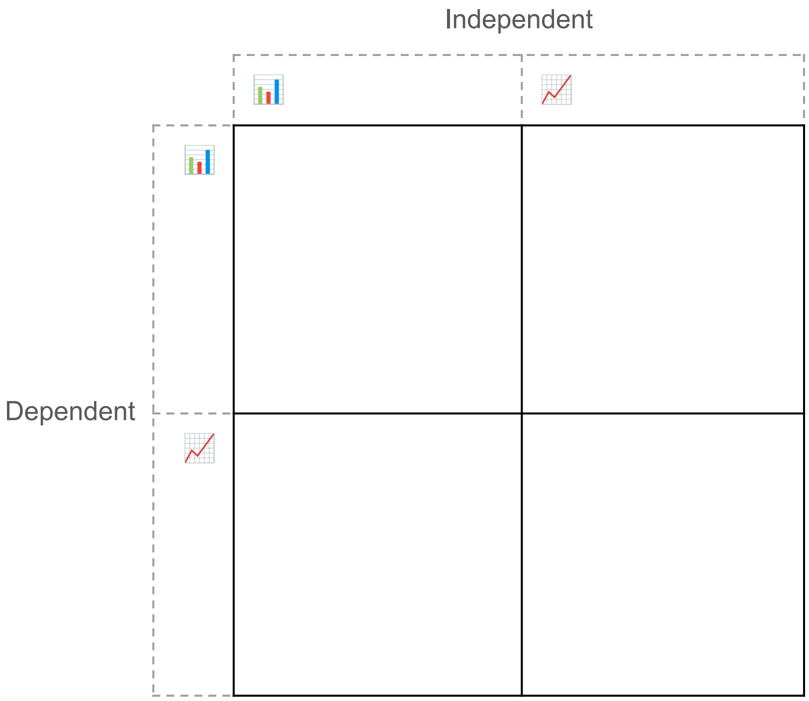
Me

temperature 📈 — sales 📈

Conceptual model | Relationships between variables

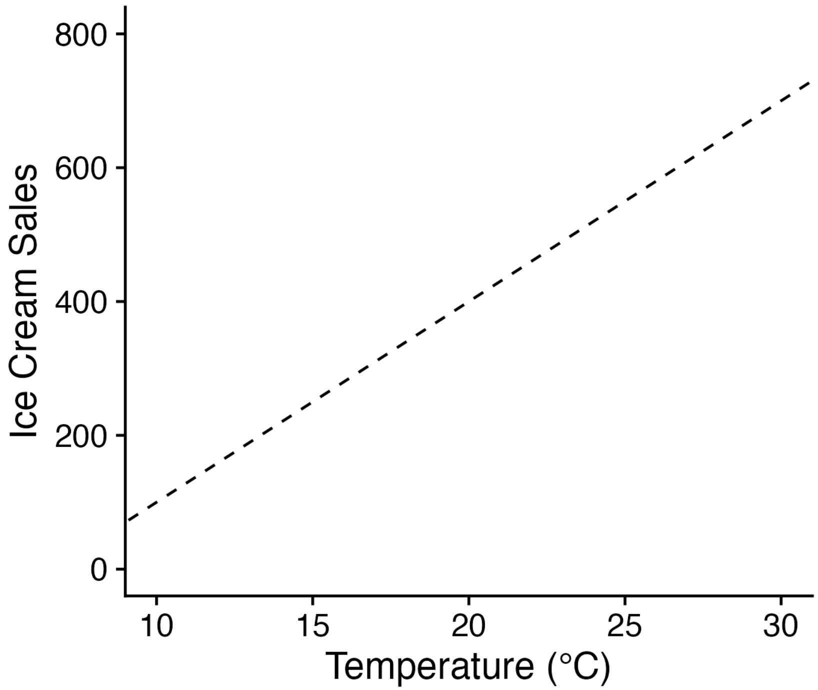
independent — dependent

predictor — outcome
explanatory — response
exogenous — endogenous
feature — target
x — y



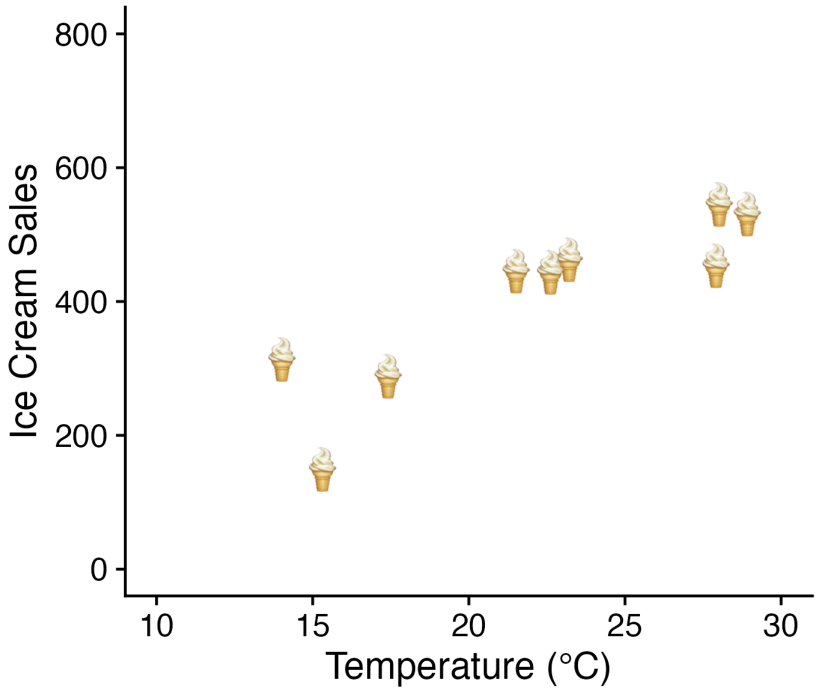
temperature 📈 — sales 📈

Functional form | Linear relationships between variables



Sales = -200 + 30 Temperature

Data collection

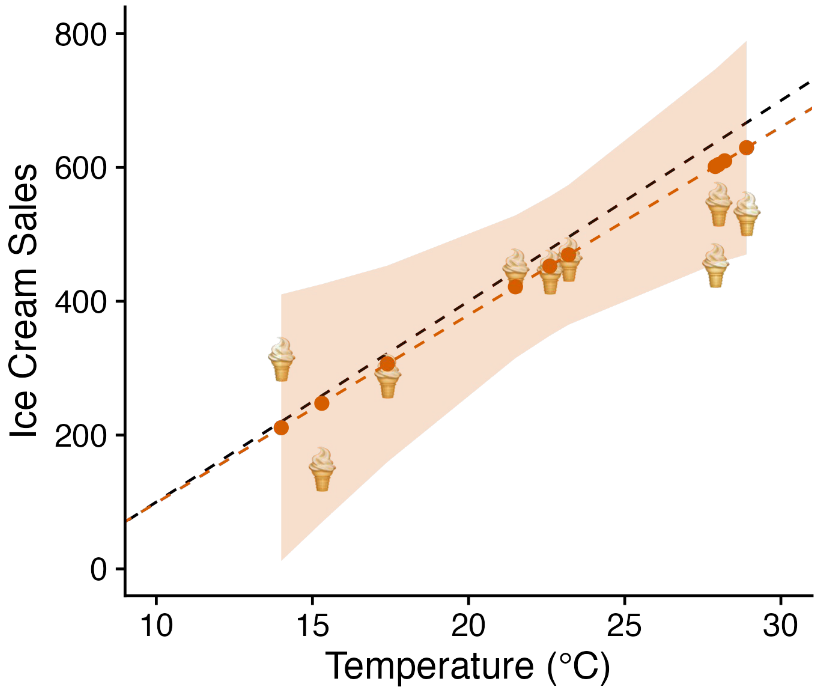



```r
ice_cream_data <- tibble::tribble(
  ~temperature, ~ice_cream_sales,
  27.9,          452,
  15.3,          148,
  17.4,          287,
  21.5,          444,
  28.2,          935,
  14,            312,
  28,            544,
  28.9,          529,
  23.2,          461,
  22.6,          442)
```

$$\text{ice\_cream\_sales} = \beta_0 + \beta_1(\text{temperature}) + \epsilon$$

```r
mod <- ice_cream_sales ~ temperature# y ~ x
mod <- ice_cream_sales ~ 1 + temperature# explicit
intercept
```

Modeling | Linear regression




```
fit <- lm(mod, data = ice_cream_data)
summary(fit); confint(fit, level = .95)
```

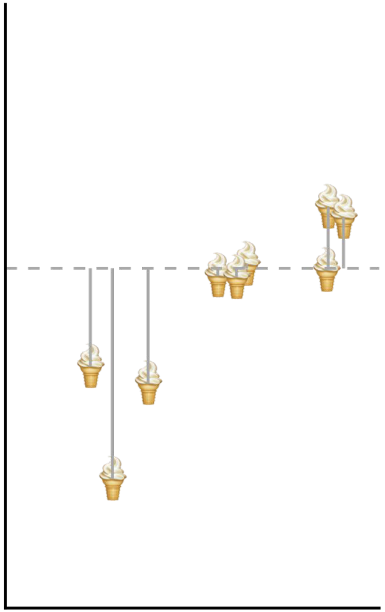
```
Call:
lm(formula = model, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-149.46  -89.73  -15.06   14.62  325.12 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)  
(Intercept) -182.199    197.447  -0.923   0.3831  
temperature   28.088      8.468   3.317   0.0106 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 142.7 on 8 degrees of freedom
Multiple R-squared:  0.579,	Adjusted R-squared:  0.5264 
F-statistic:    11 on 1 and 8 DF,  p-value: 0.01059
```

For every °C increase, sales increase on average with 9 to 48 ice creams.

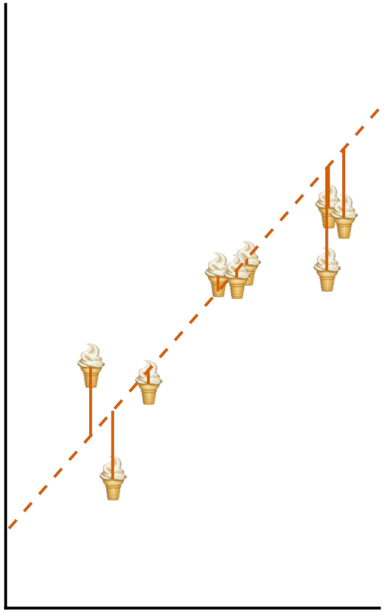
$$\widehat{\text{ice\_cream\_sales}} = -182.2 + 28.09(\text{temperature})$$

Modeling | Ordinary least squares

Residual SS (β_1 = 0)



Residual SS (β_1 = 28



Objective: minimize residual sums of squares (RSS)

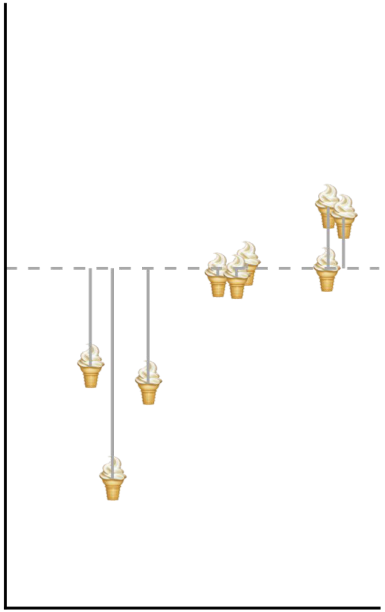
Baseline RSS $Sales_i = \beta_o + e_i$: 387052.4

Model RSS $Sales_i = \beta_o + \beta_1\ Temperature_i + e_i$: 162946

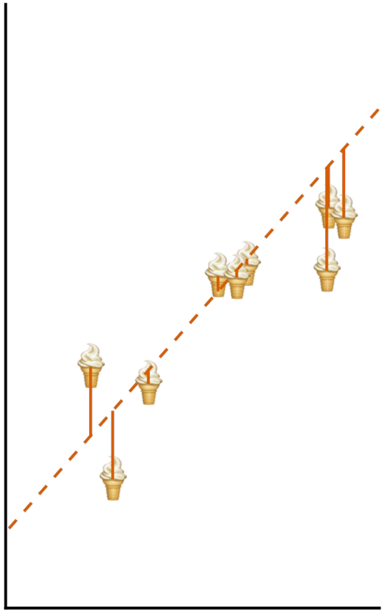
```
mod_baseline <- ice_cream_sales ~ 1
fit_baseline <- lm(mod_baseline, data = data)
summary(fit_baseline)

deviance(fit)
deviance(fit_baseline)
```

Evaluation | How well does the model describe the data?

Residual SS (β_1 = 0)



Residual SS (β_1 = 28



On average, how far off is my regression estimate from the observed values?

Standard deviation of the residuals, or root mean squared error (RMSE): √*(RSS / (n - K)*

Model RMSE: 142.717 ice creams off

Is this good or bad?

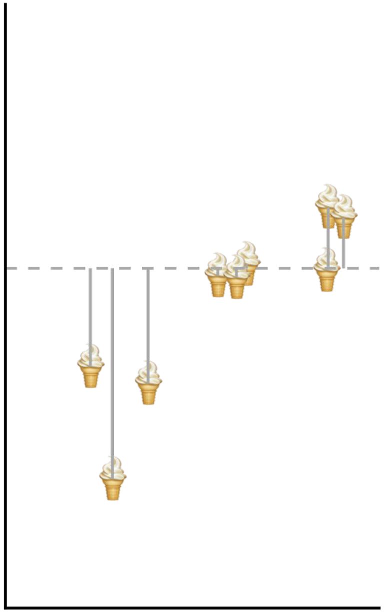
Baseline RMSE: 207.379

Still, is it good or bad? Prediction!

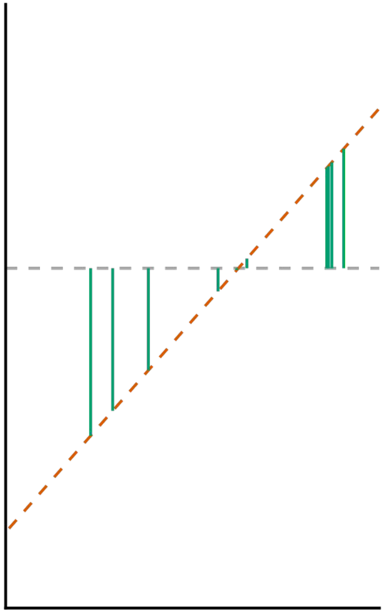
```
summary(fit)$sigma
summary(fit_baseline)$sigma
```

Evaluation | How well does the model describe the data?

Total SS



Explained SS



What is the proportion of variance shared by the data and the model? (‘verklaard’ only if experimental!)

Coefficient of determination (R^2), or ‘explained’ variance: ESS / TSS

Model R^2: 0.579

58% of variation in sales is shared with the model (i.e., temperature)

R^2 = 1 (model fits data perfectly)
R^2 = 0 (model doesn't do better than mean)

```
summary_fit$r.squared
```

📖 7.2.2

Evaluation | Is the relationship statistically significant?

Global significance (full model): *F* test

$R^2 = 0.579$ (fitted)

H_0: $R^2 = 0$
H_A: $R^2 > 0$

See next lectures.

Local significance (one coefficient): *t* value

$\beta_1 = 28.088$ (fitted)

H_0: $\beta_1 = 0$
H_A: $\beta_1 > 0$

See simulation superpower.

💡Compute *t*-statistic for β_1 (same procedure as for the mean): $t = (28.088 - 0) / 8.468 = 3.317$

Evaluation | What is the size of the effect?

Global effect size (full model)

R^2 (correlation family, variance explained)

- ❏ *$R^2 \geq .02$ weak; .13 moderate; .26 substantial; Cohen (1988)*

Cohen's f^2 (correlation family, variance explained)

- ❏ *$f^2 = R^2 / (1 - R^2)$ = explained / unexplained*
- ❏ *$f^2 \geq .02$ small; .15 medium; .35 large; Cohen (1988)*

Local effect size (one coefficient)

Cohen's f^2 (correlation family, variance explained)

- ❏ *$f^2 = (R^2_{AB} - R^2_{A}) / (1 - R^2_{AB})$*
- ❏ *No known rules of thumb*

Rules of 👍: arbitrary conventions when no substantiated rules exist.

📖 not in book

Evaluation | Sign, size, significance, smelt

💡 For every °C increase, sales increase on average with 9 to 48 ice creams.

💡 On average, the model estimate is 142.717 ice creams off from the observed values.

💡 58% of variation in sales can be attributed to the model; $F(1, 8) = 11.003$, $p = .011$ (alpha = .05).

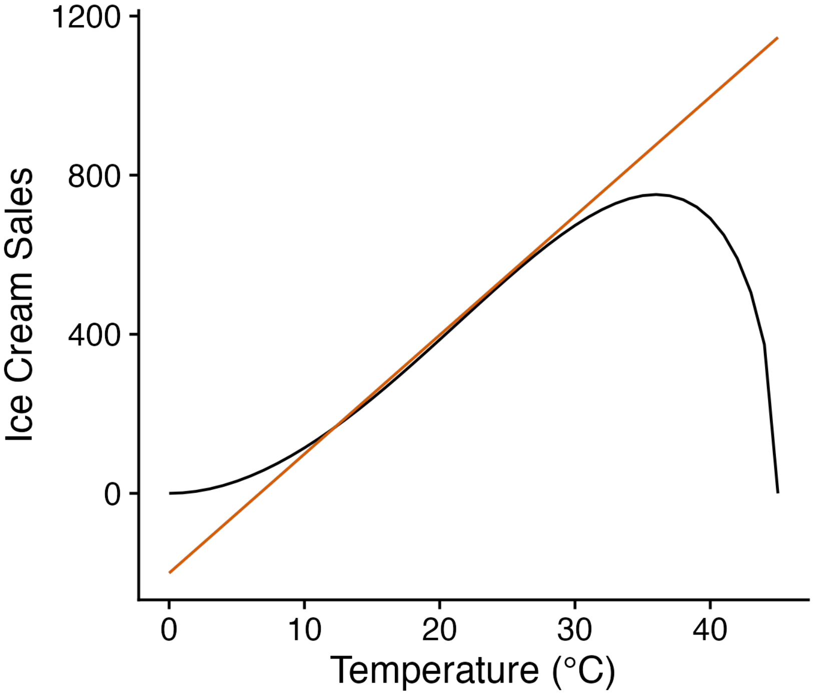
💡 The slope of temperature differed significantly from zero; $t(8) = 3.50$, $p = .011$ (two-tailed, alpha = .05).

💡 Effect size f^2 is 1.375, which is considered large (Cohen, 1988).

Prediction

- ❏ Linearity assumption
- ❏ Evaluate model on new data

```
new_data <- tibble(temperature = c(-10, 0, 10, 20,
30, 100))
predict(fit, newdata = new_data)
```



📖 7.2.4

Model thinking

What is the relationship between temperature and ice cream sales?

Conceptual model

temperature 📈 —— sales 📈

Linear regression equation

$$\text{ice\_cream\_sales} = \beta_0 + \beta_1(\text{temperature}) + \epsilon$$

R formula

```
mod <- ice_cream_sales ~ temperature
```

Introduction to R

Curled by Danielle Navarro (jasmines package, diy) 🎨

r_basics.R

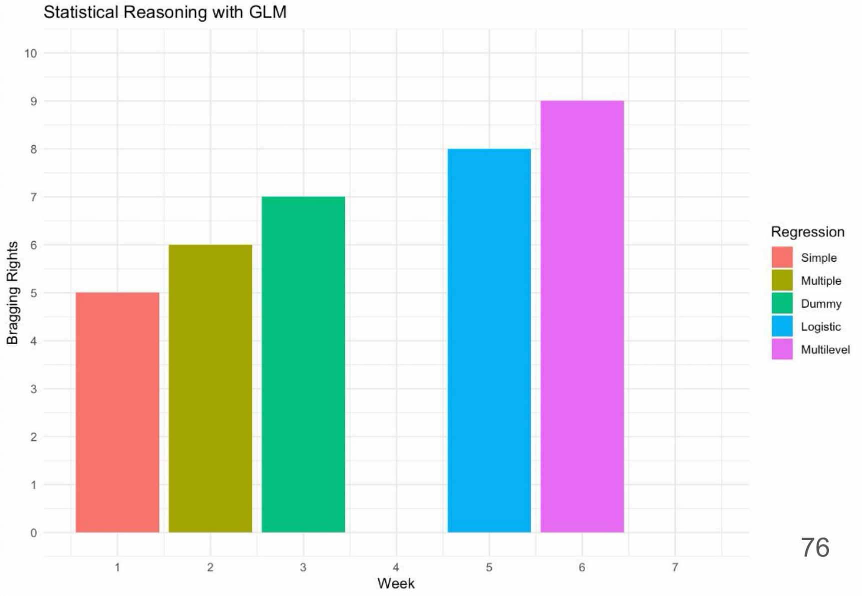
```r
# Packages ---------------------------------------------------------------

library("tidyverse")

# Data types -------------------------------------------------------------

numeric    <- 10                                  # continuous
integer    <- 10L                                 # continuous
character  <- "Statistical Reasoning"             # categorical
logical    <- TRUE                                # categorical
missing    <- NA

# Data structures --------------------------------------------------------

numeric_vector    <- c(5, 6, 7, NA, 8, 9, NA)
character_vector <- c("Simple", "Multiple", "Dummy",
                      NA, "Logistic", "Multilevel", NA)

factor <- factor(character_vector,
                 levels = c("Simple", "Multiple", "Dummy",
                            "Logistic", "Multilevel"))

data_frame <- tibble(
  week       = 1:7,
  regression = factor,
  level      = numeric_vector
)

# Plot -------------------------------------------------------------------

ggplot(data_frame, aes(x = week, y = level, fill = regression)) +
  geom_col() +
  scale_x_continuous(breaks = 1:7) +
  scale_y_continuous(breaks = 0:10, limits = c(0, 10)) +
  scale_fill_discrete(na.translate = FALSE) +
  labs(
    title = "Statistical Reasoning with GLM",
    x     = "Week",
    y     = "Bragging Rights",
    fill  = "Regression"
  ) +
  theme_minimal()
```

Environment

Data	
data_frame	7 obs. of 3 variables
Values	
character	"Statistical Reasoning"
character_vector	chr [1:7] "Simple" "Multiple" "Dummy" NA "Logistic" "Multilevel" NA
factor	Factor w/ 5 levels "Simple","Multiple",..: 1 2 3 NA 4 5 NA
integer	10L
logical	TRUE
missing	NA
numeric	10
numeric_vector	num [1:7] 5 6 7 NA 8 9 NA

Console — R 4.6.1 · ~/Downloads/statistics_class/

```
> data_frame |>
+   drop_na() |>
+   select(regression)
# A tibble: 5 × 1
  regression
  <fct>
1 Simple
2 Multiple
3 Dummy
4 Logistic
5 Multilevel
```



Best practices

- Create a *new project* for this course
- Use scripts and save regularly
  - .R
  - .Rmd (R Markdown)
  - .qmd (Quarto)
- Use *comments* excessively
- Clear history
- Understand pipes
  - `function(function(x))`
  - `x |> function() |> function()`
  - `x %>% function() %>% function()`

Help with installing packages? Setting up environment, also see book

Remind of script that comes with the book.

RStudio®

Cooling Down

Curled by Danielle Navarro (jasmines package, diy) 🎨

🔮 What's next

🧑‍💻 Weekly assignment

I. Attempt problem independently
   - Consult official resources (literature, lectures)
   - This is what will be examined

II. Consult fellow students or teachers
   - Collaborate and discuss with peers
   - Attend lecture & ask questions
   - Join the weekly drop-in consultation
   - Active engagement will increase your success

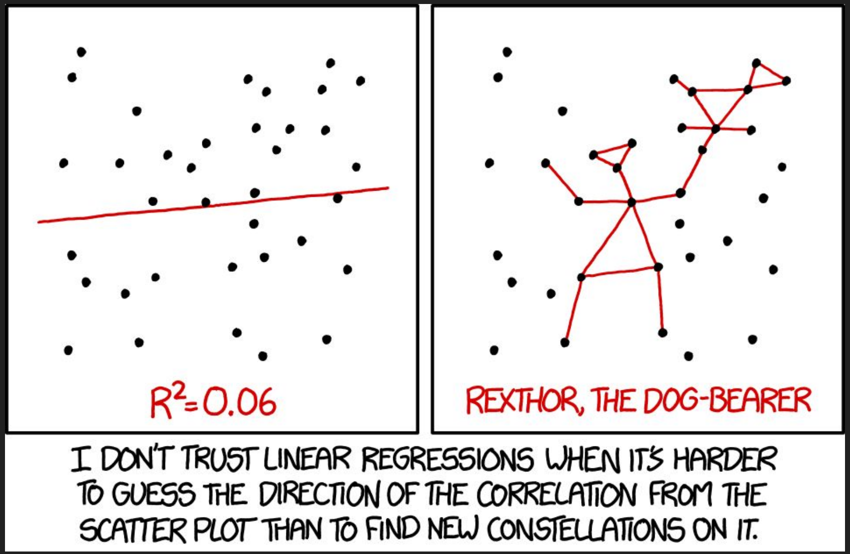
III. Use [UvA AI chat](), responsibly
   - Previous students found answers confusing
   - Only if I & II are satisfied
   - Treat as a coach, not as a solver
   - Verify and reflect, beware of plagiarism
   - Prediction ≠ intelligence

📆 Next week

⚠️ Book p. 191/2: R should be R^2

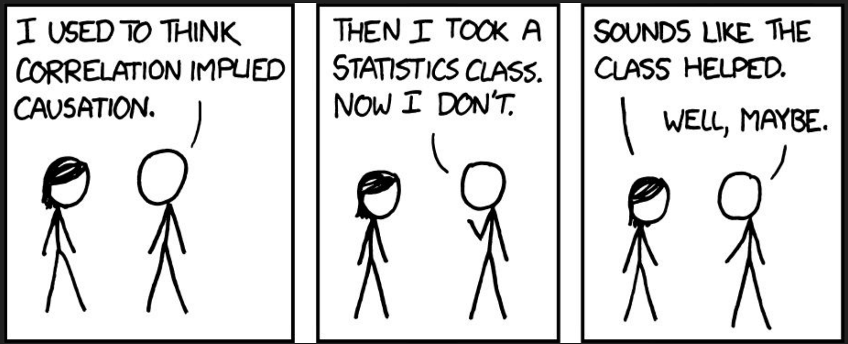

CROSS DO NOT CROSS DO NOT CROSS DO NOT CROSS DO NOT CROSS DO NOT CROSS DO NOT CROSS DO NOT CROSS DO NOT
CAUTION CAUTION CAUTION CAUTION CAUTION CAUTION

🤔 Get it? | Correlation



🧑‍🎨 Randall Munroe (wtf)

🤔 Get it? | Correlation does not imply causation

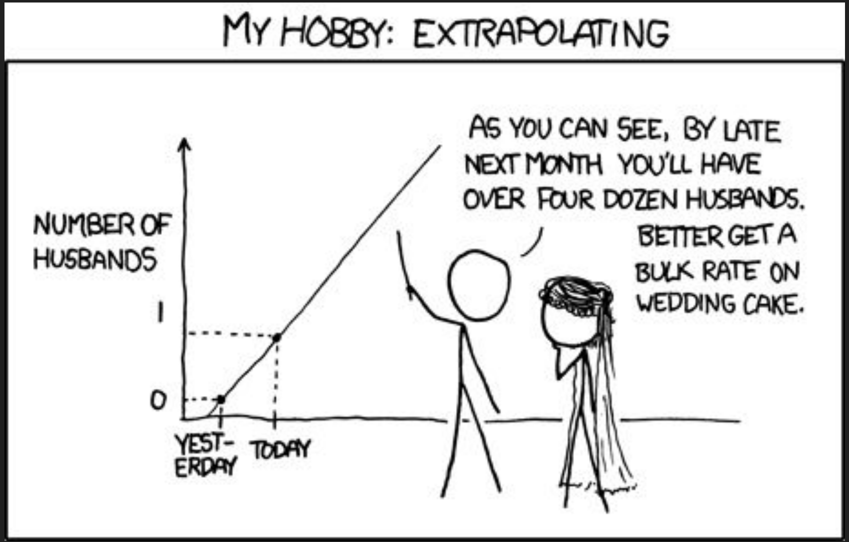


🧑‍🎨 Timo Elliot

🧑‍🎨 Randall Munroe (wtf)

🤔 Get it? | Extrapolation

If she loves you more each and every day,
by linear regression she hated you before you met.



🧑‍🎨 [Zach Weinersmith]

🧑‍🎨 [Randall Munroe] ([wtf])

✍️ Colophon

Slides

alexandersavi.nl/teaching/

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